

| Education           |                        |                                                |         |
|---------------------|------------------------|------------------------------------------------|---------|
| Year                | Degree/Certificate     | Institute                                      | CPI/%   |
| Oct 2022 – Aug 2026 | B.Tech in CS & IT      | Ajay Kumar Garg Engineering College, Delhi NCR | 7.25/10 |
| 2021 – 2022         | Senior Secondary (XII) | New Era School, Delhi NCR                      | 91.2%   |
| 2017 – 2020         | High School (X)        | SJN International School, Coimbatore           | 92.6%   |

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (May 2025 – Sept 2025)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4

*P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- Validation & Optimization:** Engineered a Jinja2-based P4Runtime testing pipeline (Tofino/Mininet) improving deployment reliability by 30%, and optimized SID recirculation to achieve 10 Gbps line-rate throughput with 20% lower inference latency.

Open Source Contributions

- WasmEdge/WasmEdge:** [#4470](#) fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch; [#4466](#) feat(docker): add standalone alpine-base image build pipeline.
- Project-HAMi/HAMi:** [#2292](#) fix(device): use len(devices) for GenReason in kunlun and awsneuron; [#2307](#) fix(device-plugin): release node lock when bind-phase patch fails; [#2259](#) fix(scheduler): handle ResourceQuota tombstones; [#2204](#) fix(metrics): prevent invalid memory ratios for unknown total memory; [#2221](#) fix(workflow): skip non-translatable issue comments.
- facebook/bpfilter:** [#507](#) build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

Cutable | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (Project Link)

Objective:

Generate editable React applications from text and visual requirements.

Approach:

Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.

Impact:

Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (Project Link)

Objective:

Determine whether a startup idea already exists by comparing it against more than 500,000 company records.

Approach:

Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.

Impact:

Raised USD 10,000 and serves more than 100 users worldwide.

Slix | *React, Go, Azure OpenAI, REST API* (Project Link)

Objective:

Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.

Approach:

Built the frontend in React and the service layer in Go; added Azure OpenAI sentiment analysis for curator reviews.

Impact:

Gives viewers a concise representation of a movie’s tone before watching.

Myprod | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* (Project Link)

Objective:

Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

Approach:

Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking; added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.

Impact:

Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

XNIC v1 | *Linux Kernel, C, PCI, BAR0 MMIO, DMA, Descriptor Rings, e1000, Intel 82540EM, net\_device, NAPI, IRQ, MSI, ethtool, QEMU, HVF, KFENCE, sparse, DPDK, rte\_ethdev, net\_pcap PMD, tcpdump, Wireshark, W5500, SPI, Device Tree, Bash, GitHub Actions, Vercel* (Project Link)

Objective:

Build and qualify a defensible clean-room Linux Ethernet driver with observable packet paths, ownership invariants, fault recovery, and a gated user-space DPDK forwarding path.

Approach:

Implemented PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX queue backpressure, ethtool counters, and serialized watchdog/reset recovery in C; added staged fault injection, KFENCE and sparse qualification, reproducible QEMU/HVF automation, Wireshark-compatible captures, a W5500 SPI driver path, and an rte\_ethdev L2 forwarder.

Impact:

Qualified 10,100 RX and TX ring wraps across 646,400 packets at zero loss, 1,000 interface cycles, 100 driver rebinds, and 100 reset-under-traffic cycles; validated DPDK partial-TX cleanup and signal-safe shutdown with the net\_pcap virtual PMD.

Technical Skills

- Language:** Python, C, C++, JavaScript, TypeScript, Go, P4
- Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86\_64, ARM64 & AMD64)
- Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

Technical Writer, Cloud Native Community Group, New Delhi (May 2024 – Dec 2024)

- Led a team of 6+ content writers and 4 designers to deliver documentation and event management support.
- Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and X, achieving a 60% bump in engagement.

- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (Nov 2023 – Nov 2025)
  - Co-led **10+** tech events & anchored the GDSC WOW flagship event at IIIT Delhi, with the **31+** crew members.
  - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

**Publications**

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” *2026 International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. **2026**.  
[IEEE Xplore](#)  [DOI: 10.1109/ICCSC67078.2026.11468597](#)

**Certifications & Honors**

|                 |                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certifications  | • <b>Open Science 101</b> – NASA    • <b>Data Science for Engineers</b> – NPTEL    • <b>Enhancing Soft Skills &amp; Personality</b> – NPTEL    • <b>IIRS Satellite Remote Sensing</b> – ISRO                                               |
| Honors & Awards | • <b>2x AKTU State Men Singles TT Gold (2023, 2024)</b> • <b>International Robotics Olympiad Bronze, Thailand (2019)</b> • <b>Merit Certification at Carnic Model United Nation (2021)</b> • <b>NCC Cadet, National Cadet Corps (2019)</b> |